Search for Σ^0 production in $\bar{\nu}_\mu$ charged-current interactions in the NuMI beam

Two selections and the expected sensitivity

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$\bar{\nu}_\mu\,\mathrm{Ar} \to \mu^+\Sigma^0 X,\ \Sigma^0 \to \Lambda\gamma,\ \Lambda \to p\pi^-$ • RHC, 1.15×10^{21} POT • signal region blind

Details: `report/other_notes/sigma0_note.pdf`

Summary

- Two independent selections:
  - **PeLEE BDT**: standard PeLEE ntuples, cross-validated gradient BDT. Has data, EXT and dirt. $S = 2.11$ / 13.3 (NuWro / GENIE), $B = 382$.
  - **Λ-style**: HyperonProduction ntuples, fitted Λ decay vertex and a wire-level "island" test. $S = 0.142$ / 0.84, $B = 4.4$: 6× the purity, 1/15 of the signal. No data or dirt in that format.
- **The result will be an upper limit.** Median discovery significance at the predicted rate: $\leq 0.6\sigma$.
- Best median expected 90% CL limit: $\sim 3\times$ the GENIE rate, $\sim 18\times$ the NuWro rate. The generators differ by 6×: NuWro has hyperon FSI, GENIE does not.
- Under one signal model the **PeLEE chain, its cut set for the background systematic, gives the better limit** under both generators up to 30%. Purity alone pointed the other way.
- **New GENIE Σ^0 QE sample** (250k events): agrees with the overlay's Σ^0 to 2%; the old GENIE hyperon sample held only 754 distinct interactions.
- Next: the background systematics of the PeLEE chain, which set the limit.

The signal

- $\bar{\nu}_\mu$ CC, a Σ^0 in the final state, true vertex in the FV.
- $\Sigma^0 \to \Lambda\gamma$ ($\sim 100\%$): soft photon, $E_\gamma = 74.5$ MeV in the rest frame.
- $\Lambda \to p\pi^-$ (63.9%): the Λ is slow ($\beta\gamma \approx 0.4$), flight median 1.85 cm, mean 3.2 cm.
- Primary-vertex resolution 5.5 cm: the flight cannot be measured from the Pandora vertex ($r = -0.014$), but the decay topology can be tested at wire level.
- Reconstruction ceiling: all four particles found in 18% of signal; the π^- in 47%.

In-FV Σ^0 per 1.15×10^{21} POT

GENIE overlay (no hyperon FSI)	83.3
GENIE Σ^0 QE sample (new)	81.9
GENIE hyperon sample (old)	110.1
NuWro run3b, flux-weighted	12.8

NuWro/GENIE $= 0.15$: hyperon FSI 4.19×, old flux 1.48×, production 1.05×; product 6.51× = measured. A NuWro CCQE control run agrees with GENIE's rate to 17%.

A model choice, not a bug: every result is quoted under both.

Samples

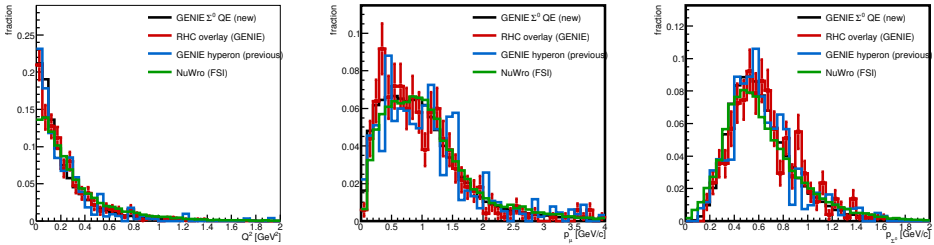
format	sample	content	scale
PeLEE	Σ^0 signal	NuWro run3b overlay (local production), 250k events, flux-weighted	8.6×10^{-4}
	ν background	RHC overlay Runs 1, 2, 3, 4a, 4b, 4c, 1.55×10^{22} POT	0.075
	dirt	RHC Runs 1, 2b, 3b, 4a, 4b, 1.74×10^{21} POT, old flux	0.66
	EXT	run3b beam-off, 3.27×10^7 gates	0.75
	GENIE Σ^0, fake data	GENIE Σ^0 QE overlay (local), 251k events	9.4×10^{-4}
hypana	Σ^0 signal	the same GENIE QE and NuWro run3b productions	
	ν background	14 164 files, 7.05×10^{21} POT	0.16
	EXT	16 570 events (local reco2 + HyperonProduction)	150
	dirt, data	—	

Fixed in the PeLEE background (17 October):

- the dirt was FHC ($\langle E_\nu \rangle$ and spectra), now RHC; it is on the old NuMI flux and takes the NuWro flux map;
- the Run 3 overlay was missing (+55% background statistics);
- EXT added; GENIE Σ^0 inside the overlay no longer counted as background.

New GENIE Σ^0 QE sample (2026-10-30)

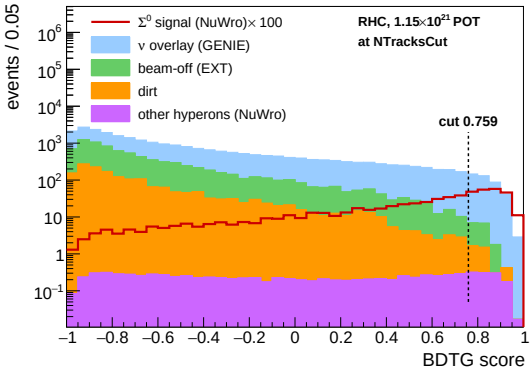
GENIE `LambdaCCQE`, primary Σ^0 selected, official dk2nu RHC flux, our overlay chain on run3b EXT: 2506 jobs, 250 555 events, 1.23×10^{24} POT.



- In-FV Σ^0 per 1.15×10^{21} POT: **81.9** (QE sample), 83.4 ± 3.7 (overlay), 110.1 (old hyperon sample). Shapes match the overlay.
- With the overlay's PPFX CV weight (mapped in E_ν): agrees with the overlay's own Σ^0 to 2% at every PeLEE cut, after the BDT too.
- The old GENIE hyperon sample holds only 754 distinct interactions, each overlaid $\sim 20\times$: spiky spectra, $1.32\times$ the rate, a harder p_μ. Everything that used it is redone.
- Overlay non-QE Σ^0 come with a kaon (vetoed): the QE sample is the full GENIE prediction.

PeLEE chain: selection and BDT

- Cosmic veto (topological score > 0.2), 1–3 showers, muon candidate, < 5 tracks.
- Proton and pion: best *ordered pair* from a joint per-role score; pion purity 33.5 $\rightarrow$ 42.5% (ceiling 51%). Formed when present, *not required*: requiring them halved the figure of merit.
- 57 BDT inputs: PID, distances, angles, Λ and Σ^0 masses, pointing, π^0 and second-shower variables, `HasPair`.
- Two-fold cross-validation by event parity: every score is out of sample.
- **Retrained 18 Oct**: EXT in the background tree (+5%), and training on 95% of each fold, not TMVA's default 50% (+4%). AUC 0.889 $\rightarrow$ 0.900.



cross-validated BDTG score at preselection

PeLEE chain: results at BDTG > 0.759

stage	raw sig.	eff.	S	B
MuonCandidate	11107	51.6%	6.47	28978
NTracksCut	10931	50.8%	6.37	26549
BDTG	**3644**	**16.9%**	**2.11**	**382**

B at BDTG	old BDT (0.616)	new (0.759)	raw
overlay	715.7	367.9	5178
EXT	109.5	11.3	15
dirt	13.8	2.3	7
other hyperons	1.7	1.1	1841
total	840.7	**382.4**	

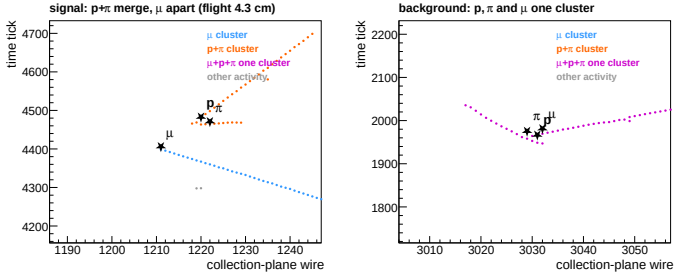
purity 0.34% → **0.55%**, $S/\sqrt{S+B}$ 0.098 → **0.108**

GENIE signal through the same chain
GENIE Σ^0 QE sample (never trained on): $S_{\mathrm{GENIE}} = 13.32 \pm 0.11$ (15 142 raw); the overlay's own Σ^0: 13.1 ± 1.0 (187 raw). Efficiency 18.1% against NuWro's 16.8%.
$\Rightarrow$ the limit on the number of events is model-independent to 7%.

Detector variations (Run 3 set, 9 variations)
None resolved: quadrature 5.2% on B (MC noise 6.8%), 21.7% on S (noise 21.1%).
$\Rightarrow \lesssim 7\%$ on B; not measured on S.

Λ-style chain: following the Λ CCQE analysis

- The Λ CCQE analysis reaches 91% purity; at equal efficiency our earlier hypana BDT had 2.8%.
- Score every ordered track pair; pair precuts are a preference, not a veto.
- Fit the decay vertex (V fit): decay length, pointing angle α.
- **Island test**: grow clusters of raw wire activity from the μ, p and π starts; require p and π to merge and the μ to stay apart, on ≥ 2 planes.
- Needs wire windows: hypana format only.



collection plane; left: signal with 4.3 cm flight, right: background

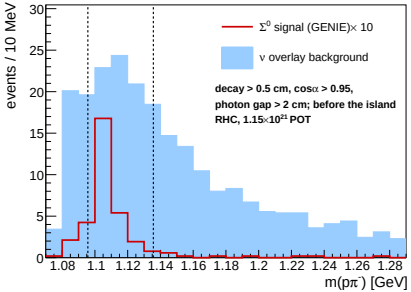
Λ-style chain: operating point

stage	raw sig	raw bkg	S (GENIE)	B
MuonCandidate	3043	123000	58.7	20113
pair	921	51238	17.8	8378
$m_{p\pi}$ ±20 MeV	504	16391	9.72	2680
pointing	144	706	2.78	115.4
photon gap > 2 cm	139	549	2.68	89.8
island, ≥ 2 planes	**69**	**23**	**1.33**	**3.76**

+ EXT 0.62 ± 0.27 + NuWro FSI $0.02 \Rightarrow B = 4.40$.
S column: old GENIE hyperon sample. GENIE QE sample: $S = 0.896 \pm 0.029$ (956 raw), purity 17%. NuWro: $S = 0.142$ (245 raw).

Island pass rate vs true flight

flight	1–2	2–3	3–5	5–10 cm
connectivity 3×40	19%	45%	70%	86%
2×20 (now)	43%	80%	85%	91%



before the mass window and the island test

Out of sample: the photon gap, fixed in September on the Run 3 list, halves the background of the independent extension at 96% signal efficiency.

Λ-style chain: beam-off background

- 3 349 run3b `ext_unbiased` files through reco2 and HyperonProduction locally: 16 570 events, 150 data events each. Too few to reach the operating point (0 of 76 pass the pointing cut).
- Factorised: EXT at MuonCandidate $\times$ pass fraction of the overlay's own cosmic-dominated candidates.

step	EXT raw	EXT P	cosmic MC raw	cosmic MC P	all-bkg P
MuonCandidate	76	–	4 220	–	–
pair	8	0.105 ± 0.035	571	0.135 ± 0.005	0.42
$m_{p\pi}$ window	3	0.38 ± 0.17	165	0.289 ± 0.019	0.32
pointing	0	0 of 3	7	0.042 ± 0.016	0.043

- **EXT** $= 0.62 \pm 0.27$. Real cosmics confirm the factorisation where they have events: 8 of 76 pairs vs 10 from the cosmic MC, 32 from the neutrino rate.
- Open: the island and gap steps borrow the neutrino-background rate; without that, EXT < 6.2 at 90% CL.

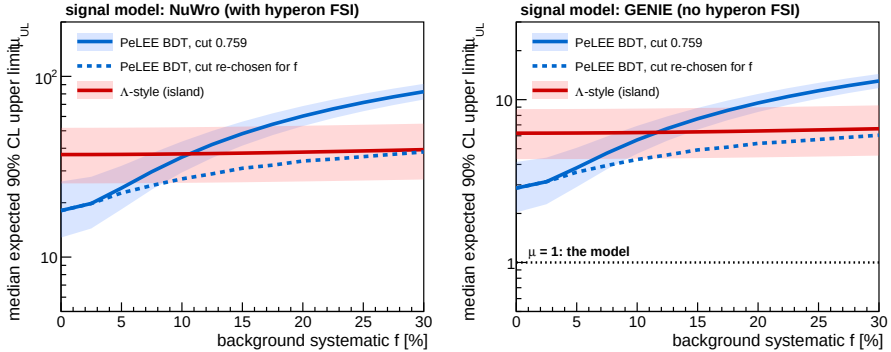
Expected sensitivity: method

- One-bin counting, $n \sim \mathrm{Pois}(\mu Se + b)$, with μ the Σ^0 rate in units of the model.
- b: lognormal, width = MC statistics $\oplus$ background systematic f (scanned 0–30%).
- e: lognormal, 20% signal normalisation (placeholder) $\oplus$ signal MC statistics.
- Bayesian limit, flat prior in μ, nuisances marginalised; hybrid CL_s is 4% tighter. Both reproduce the PDG table for a known background.
- Expected limit: at the median n of $\mathrm{Pois}(B)$, band from the 16 and 84% quantiles.
- The PeLEE BDT cut is also re-chosen for each f to minimise the limit.
- `Sigma/pelee/Sensitivity.C`

Both chains under one model

NuWro: PeLEE $S = 2.11$, Λ-style $S = 0.142$. GENIE (QE sample, PPFX-weighted): PeLEE 13.3, Λ-style 0.84.
Under the same model, purity favours the Λ-style chain $\sim 6\times$ (3.1% vs 0.55%, NuWro) and signal yield favours the PeLEE chain $15\times$.

Expected sensitivity: result



median μ_{UL}, NuWro / GENIE	$f = 0$	$f = 15\%$	$f = 30\%$
PeLEE, cut 0.759	18.1 / 2.86	48.2 / 7.63	82.3 / 13.0
PeLEE, cut re-chosen	18.1 / 2.86 (0.750)	31.0 / 4.91 (0.890)	38.2 / 6.05 (0.925)
Λ-style	36.9 / 6.22	37.6 / 6.34	39.3 / 6.63

What the sensitivity says

1. **A limit, not an observation.** Best median $Z = 0.63\sigma$ (GENIE). The best limit is ~ 220 in-FV Σ^0 events; GENIE predicts 73, NuWro 13.
2. **The PeLEE chain is better under both generators up to** $f = 30\%$: $2.0\times$ at $f = 0$, $1.4\times$ at 10%, $1.2\times$ at 15% (NuWro; GENIE $2.2\times$, $1.5\times$, $1.3\times$). At 30%: 38.2 vs 39.3 and 6.05 vs 6.63.
3. **The Λ-style limit is set by Poisson counting** at $B \approx 4.4$: perfect knowledge of B moves it by 3%, dirt at its bound by at most 16%. Hypana data and dirt would give a cross-check with different systematics, not a better limit.
4. **The background systematic drives the PeLEE limit**: $\times 1.7$ from $f = 0$ to 15% even with the cut re-chosen.
5. **Choose the BDT cut from the systematic budget**: at $f = 10\%$, 0.875 instead of 0.759 is 25% better. $S/\sqrt{S + B + (fB)^2}$ agrees up to 10% and then picks cuts with too few raw events.

Next steps

1. **Background systematics, PeLEE chain**: flux (`weightsPPFX`), cross section (`weightsGenie`) and reinteraction (`weightsReint`) multisims at one bin – the ntuples carry them, no framework port needed – plus detector variations ($\lesssim 7\%$).
2. **Constrain B from data**: fit the low-BDT-score region, where the signal is negligible, and extrapolate; the π^0 normalisation is the target.
3. **Set the BDT cut** from the resulting f; redo the detector variations there.
4. **Signal model**: quote under NuWro and GENIE. Redo the Λ cross-check and the island tuning on the GENIE QE sample; try BDT training with GENIE signal.
5. **Closure**: Poisson-thrown fake data, with the GENIE Σ^0 QE sample as an injected alternative signal.
6. **Unblind** the RHC beam-on PeLEE data and set the observed limit.
7. Later: Λ-style data and dirt as a cross-check; FHC+RHC.

Backup

Backup: BDT retraining, out of sample

training	AUC	B/B_0 at signal efficiency 50%	40%	20%	peak $S/\sqrt{S+B}$ value	cut
15 Oct (no EXT, half fold)	0.8894	0.0412	0.0253	0.0067	0.0993	0.690
EXT, half fold (3 seeds)	0.896	0.038	0.023	0.006	0.104–0.105	
EXT, 95% of fold	**0.9000**	**0.0358**	**0.0215**	**0.0061**	**0.1077**	**0.759**
EXT, 95% (2 other seeds)	0.900–0.902	0.035	0.021	0.006	0.108–0.109	
inputs clamped (2 seeds)	0.904	0.034	0.021	0.006	0.110	

- EXT in training: at 50% signal efficiency 0.9% of EXT survives against 2.2%; overlay unchanged.
- Full fold: every full-fold seed beats every half-fold seed; reseeding moves the figure of merit by $\pm 0.5\%$. The BDT still gains from more signal.
- Default seed adopted, not the best one; `TrainBDT.C` reproduces it exactly.
- Not adopted yet: some inputs carry $\pm$FLT_MAX or garbage, which defeats TMVA's even cut grid. Clamping them gives -4% background at 50% efficiency, $+1.6\%$ FoM; it goes in at the next retraining.

Backup: detector variations at BDTG > 0.759

variation	S var/CV	B var/CV
LY attenuation	1.000 ± 0.000	0.997 ± 0.005
LY down	0.975 ± 0.025	0.985 ± 0.006
LY Rayleigh	0.980 ± 0.039	0.987 ± 0.008
recombination	1.062 ± 0.088	0.973 ± 0.030
SCE	0.921 ± 0.090	0.998 ± 0.034
WireMod θ_{XZ}	0.907 ± 0.076	1.002 ± 0.023
WireMod θ_{YZ}	0.840 ± 0.078	1.026 ± 0.026
WireMod X	0.975 ± 0.075	1.027 ± 0.026
WireMod YZ	0.964 ± 0.095	0.986 ± 0.023

- Same generated events, compared event by event with the CV weight on both sides; old-flux samples, so ratios only.
- CV keeps 51 raw signal and 1 260 raw background events at this cut.
- Quadrature equals MC noise on both sides: nothing resolved.
- WireMod θ_{YZ} (2.0σ) and θ_{XZ} (1.2σ) pull the signal down, as at the previous cut.

Backup: Λ-style sidebands and handles

Sidebands (one-plane point, for statistics)

region	S	B	S/B
signal region (blind)	1.89	14.9	12.7%
mass, +40 to +200 MeV	0.02	10.5	0.2%
$0 < \cos\alpha < 0.8$	0.50	58.2	0.9%
island fails	0.79	74.9	1.1%

All RES/DIS-dominated, like the signal region. The mass sideband exists only above the peak: $m_{p\pi}$ has a floor at 1.078 GeV.

Photon gap (> 2 cm, on): 18.7% of background "photons" start at the vertex vs 3.5% of signal; 11 of the 15 events it removes at two planes are hadrons called showers.

Michel tag (off): the background muon is a wrong-sign μ^- 78% of the time and is captured in argon 75% of the time; removes 87% true μ^- but costs 22% of signal and does not help on top of the gap.

Island tuning out of sample: chosen on Run 3, scored on the extension: a plateau (0.425–0.438), the chosen point 0.004 from the top.

Backup: what moves the Λ-style limit ($f = 15\%$, NuWro)

variant	B	median n	median μ_{UL}
nominal	4.40	4	37.6
without the EXT estimate	3.78	4	39.8
dirt at its bound ($+0.33 \pm 0.33$)	4.73	5	43.5
no signal normalisation systematic	4.40	4	34.4
background known exactly	4.40	4	36.5

- The median n is an integer: a lower B at the same n gives a *higher* limit, which is why dropping EXT makes it worse. The Asimov limit ($n = B$) moves smoothly: 40.4, 38.2, 41.6, 36.9, 39.2.
- Under GENIE from the old hyperon sample instead of the QE sample: 5.27 instead of 6.34.